



ITCS457 Decision Support and Business Intelligence Systems
Project 2: Model-Based Decision Supporting Tools
(Group 2)

Group Members

Chanisara	Jonsomjid	Reg no. 6588033
Poonnapa	Kongkhum	Reg no. 6588049
Junhao	Zhang	Reg no. 6588128

Submit to

Assoc.Prof. Dr. Jarernsri Mitranont

Management Information Systems

The Faculty of Information and Communication Technology
Mahidol University

Executive Summary

This report presents a comprehensive Decision Support System (DSS) developed for Metropolia Smart Transportation to optimize public transportation operations through data-driven insights. The project demonstrates the practical application of ETL (Extract, Transform, Load) processes and business intelligence visualization techniques to address real-world operational challenges in urban transit management.

The team analyzed five key datasets: routes, vehicles, ridership, maintenance, and customer feedback using Power Query for data transformation and Excel for creating interactive dashboards. The analysis revealed critical insights across four operational dimensions: daily ridership patterns, route efficiency, fleet reliability, and customer satisfaction. Key findings indicate that high-demand routes (R04, R05, R09) experience significant delays during peak hours, with on-time performance dropping below acceptable thresholds. Route utilization analysis showed overcrowding on routes R08, R10, and R12 (>90% load factor), while routes R02, R06, and R16 remain underutilized (<40% load factor), suggesting resource misallocation.

Maintenance analysis identified buses as the highest-cost vehicle type with peak expenditures in March and August, while customer feedback revealed that 25% of comments were negative, primarily citing delays and overcrowding issues that correlate directly with operational performance metrics. The integrated dashboard provides the Chief Operating Officer (COO) with dynamic, filterable visualizations that enable immediate identification of problem areas and data-backed decision-making for resource allocation, schedule optimization, and service quality improvement.

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Introduction

Background

Urban public transportation systems serve as the backbone of modern cities, facilitating millions of daily journeys while facing increasing pressure to improve efficiency, reliability, and customer satisfaction. As cities grow and passenger demands evolve, transportation authorities must shift from reactive management to proactive, data-driven decision-making. Traditional approaches relying on manual reporting and intuition are no longer sufficient to address the complexity of modern transit operations.

Metropolia Smart Transportation, like many metropolitan transit systems, manages a diverse fleet of vehicles operating across multiple routes with varying demand patterns, maintenance requirements, and performance standards. The organization collects vast amounts of operational data daily, including ridership numbers, vehicle status, maintenance records, and customer feedback, yet struggles to transform this data into actionable insights that can guide strategic and tactical decisions.

Problem Statement

The Chief Operating Officer (COO) of Metropolia Smart Transportation faces several critical operational challenges:

1. **Service Reliability Issues:** Inconsistent on-time performance, particularly during peak hours, leading to passenger dissatisfaction and reduced system credibility
2. **Resource Allocation Inefficiencies:** Unclear understanding of which routes are over- or underserved, resulting in suboptimal vehicle deployment and operational costs
3. **Maintenance Cost Management:** Rising maintenance expenses without clear visibility into cost drivers, preventive maintenance opportunities, or vehicle lifecycle patterns
4. **Customer Satisfaction Gaps:** Limited integration between operational performance metrics and passenger feedback, making it difficult to prioritize service improvements

Without an integrated view of these interconnected factors, the COO cannot effectively identify root causes, predict future issues, or justify resource allocation decisions to stakeholders.

Project Objectives

This project aims to develop a comprehensive Decision Support System that addresses these challenges through the following objectives:

1. **Data Integration:** Consolidate disparate datasets (routes, vehicles, ridership, maintenance, feedback) into a unified analytical framework using ETL processes
2. **Performance Visualization:** Create interactive dashboards that provide real-time insights into ridership patterns, route efficiency, fleet reliability, and customer satisfaction
3. **Decision Support:** Enable data-driven decision-making by identifying operational bottlenecks, resource allocation opportunities, and service quality improvement areas
4. **Scalability:** Design a reusable methodology that can be expanded to include additional data sources and analytical dimensions in the future

Scope and Approach

The project employs Microsoft Excel and Power Query as the primary tools for data transformation and visualization, chosen for their accessibility, integration capabilities, and widespread organizational adoption. The methodology follows a systematic approach:

1. **Extract:** Load raw data from five operational datasets into Power Query
2. **Transform:** Clean, standardize, and enrich data with calculated metrics (load factor, vehicle age, sentiment scores)
3. **Load:** Create structured tables in Excel for analysis
4. **Visualize:** Develop four specialized reports focusing on ridership, route efficiency, maintenance, and customer feedback
5. **Integrate:** Combine reports into a unified dashboard with interactive filtering capabilities

This approach demonstrates how business intelligence techniques can transform operational data into strategic insights, empowering transportation managers to make faster, more accurate, and more confident decisions that improve service quality while optimizing resource utilization.

Research on the Public Transportation Operation

Overview of Public Transportation Performance Management

Public transportation systems worldwide have increasingly adopted data-driven approaches to optimize operations, improve service quality, and justify investments to stakeholders. Modern transit agencies collect extensive operational data through automated systems, including GPS tracking, automatic passenger counters (APC), automated fare collection (AFC), and computerized maintenance management systems (CMMS). This research examines industry best practices, key performance indicators (KPIs), and dashboard implementations from leading transit authorities to inform the development of Metropolia's Decision Support System.

Key Performance Indicators in Transit Operations

Based on industry standards established by organizations such as the American Public Transportation Association (APTA) and the Federal Transit Administration (FTA), transit agencies typically monitor performance across five critical dimensions:

1. Service Delivery & Reliability

- On-Time Performance (OTP): Percentage of trips departing/arriving within defined time windows (typically $\pm 2-5$ minutes). Industry benchmarks suggest 85-95% as acceptable performance [1].
- Service Regularity: Headway adherence measures consistency between scheduled services.
- Missed Trips: Percentage of scheduled trips canceled due to operational issues.

2. Ridership & Demand

- Passenger Boardings: Total passengers carried, measured daily, monthly, and annually.
- Load Factor: Ratio of actual passengers to vehicle capacity, indicating crowding levels. The optimal range is typically 60-80% during peak hours [2].
- Passenger-Miles: Total distance traveled by all passengers, measuring system utilization.

3. Operational Efficiency

- Vehicle Productivity: Passenger-miles per revenue vehicle hour.
- Fleet Utilization: Percentage of available vehicles in active service.
- Cost per Passenger Mile: Operational cost efficiency metric.
- Farebox Recovery Ratio: Revenue generated relative to operating costs.

4. Fleet Management & Maintenance

- Mean Distance Between Failures (MDBF): Average miles traveled before mechanical failures.
- Maintenance Cost per Vehicle: Average annual or monthly maintenance expenditure.
- Vehicle Availability: Percentage of fleet ready for service daily.
- Age Distribution: Fleet age profile indicating replacement needs [5].

5. Customer Experience

- Customer Satisfaction Scores: Survey-based ratings of service quality.
- Complaint Rates: Number of complaints per 1,000 passenger trips.
- Service Accessibility: Coverage and frequency metrics.

Industry Examples and Best Practices

Chicago Transit Authority (CTA) Performance Dashboard. The Chicago Transit Authority operates one of the most comprehensive public-facing performance dashboards in North America [3]. Their system provides:

Key Features:

- Real-Time Updates: Daily updates of ridership, on-time performance, and service metrics
- Historical Trending: Multi-year comparisons showing performance evolution
- Route-Level Granularity: Individual performance metrics for each bus and rail route
- Transparency Focus: Public accountability through accessible data visualization

Dashboard Components:

- Daily ridership graphs showing seasonal and weekly patterns
- On-time performance tracked against the 85% target threshold
- Service reliability indicators, including missed trips and delays
- Customer complaints categorized by type (delays, cleanliness, safety)

Findings Relevant to Our Project:

- CTA's dashboard demonstrates the importance of benchmarking actual performance against established targets (e.g., 85% OTP target)
- Route-level analysis enables the identification of specific problem areas rather than system-wide averages
- Multi-year trending helps distinguish seasonal patterns from genuine performance degradation
- Public transparency drives organizational accountability and continuous improvement

North County Transit District (NCTD) Operations Report

NCTD's FY2023 Transit Operations Performance Report [4] exemplifies comprehensive operational analysis:

Analytical Approach:

- Comparative Analysis: Year-over-year comparisons of key metrics
- Mode-Specific Reporting: Separate analysis for bus, rail, and paratransit services
- Root Cause Analysis: Investigation of performance deviations with explanatory context
- Budget Integration: Linking operational metrics to financial performance

Key Metrics Reported:

- Ridership trends with demographic and route-level breakdowns
- Vehicle revenue hours and miles
- On-time performance with penalty/reward implications
- Customer satisfaction survey results
- Maintenance cost per vehicle by mode

Findings Relevant to Our Project:

- Vehicle type (bus vs. tram) significantly affects maintenance costs and reliability
- Seasonal variations in ridership require flexible resource allocation strategies
- Integration of customer feedback with operational metrics validates performance issues
- Maintenance cost tracking by vehicle type enables better budget forecasting

Federal Transit Administration (FTA) National Transit Database

The FTA's National Transit Database (NTD) [2] provides standardized reporting frameworks used by all US transit agencies receiving federal funding:

Standardized Metrics:

- Unlinked passenger trips (total boardings)
- Passenger-miles traveled
- Vehicle revenue miles and hours
- Operating expenses by category
- Fleet inventory and condition
- Safety and security events

Data Quality Requirements:

- 95% sampling confidence for ridership data using APCs
- Monthly reporting cycles with annual certifications
- Asset condition assessments using standardized scales
- Mandatory reporting of major service changes

Findings Relevant to Our Project:

- Standardized definitions ensure consistent measurement across time periods
- Load factor calculations require accurate capacity and ridership data
- Maintenance cost categorization (preventive vs. corrective) provides insight into fleet health
- Temporal granularity (daily, monthly, annual) serves different decision-making needs

Dashboard Design Principles from Industry Leaders

Research into leading transit dashboards reveals several consistent design principles:

1. Hierarchy of Information

- Executive View: High-level KPIs with traffic-light indicators (red/yellow/green)
- Operational View: Detailed metrics for day-to-day management
- Analytical View: Trend analysis and root cause investigation tools

2. Interactivity and Filtering

- Date range selectors for temporal analysis
- Route/vehicle filters for targeted investigation
- Drill-down capabilities from system-level to route-specific detail

3. Visual Design Standards

- Consistent color coding (e.g., red for below-target performance)
- Combination charts showing related metrics (e.g., ridership volume vs. load factor)
- Minimal text with clear chart titles and legends
- Mobile-responsive design for field access

4. Actionable Insights

- Automatic alerts for metrics exceeding thresholds
- Comparative displays showing best/worst performers
- Trend indicators showing improvement or deterioration
- Direct links to the underlying data for investigation

Application to Metropolia Smart Transportation Project

The research findings directly informed our project design in several ways:

1. KPI Selection:

Our four report categories (ridership, route efficiency, maintenance, customer satisfaction) align with industry-standard performance dimensions

2. Metric Calculations:

- Load factor calculation matches APTA standards
- On-time performance measurement follows FTA guidelines
- Maintenance cost tracking by vehicle type reflects NTD requirements

3. Dashboard Structure:

- CTA's route-level granularity inspired our RouteID filtering capability
- NCTD's mode-specific analysis guided our vehicle type comparisons
- Industry interactive design principles shaped our slicer implementation

4. Integration Approach:

- Following best practices, we integrated operational metrics (ridership, OTP) with qualitative feedback (customer satisfaction)
- Multi-dimensional analysis enables root cause identification (e.g., high load factor correlating with low OTP and negative feedback)

5. Decision Support Focus:

- Research emphasized that effective dashboards must enable action, not just reporting
- Our design highlights specific problem routes (R04, R05, R09) and overcrowded services (R08, R10, R12)
- Resource reallocation opportunities are visually evident through underutilized route identification

Conclusion of Research Findings

Industry research demonstrates that successful public transportation Decision Support Systems share common characteristics: they measure standardized KPIs, provide multi-level analytical capabilities, integrate diverse data sources, and present information in visually intuitive formats that enable rapid decision-making. Leading transit agencies have moved beyond simple reporting to predictive analytics and real-time operational adjustments. While our project focuses on foundational dashboard development using Excel, the methodology and analytical framework are scalable to more advanced platforms and predictive capabilities as Metropolia's data maturity evolves.

The examples from CTA, NCTD, and FTA validate our approach while highlighting opportunities for future enhancement, including real-time data integration, predictive maintenance modeling, and mobile accessibility for field operations.

ETL Process and Data Visualization

1. Extract: Load Data into Power Query

In Excel: Go to Data → Get Data → From Workbook, and then choose the Transportation.xlsx. Next, in the Navigator window, select these 5 sheets (routes, vehicles, ridership, maintenance, feedback). Finally, click Transform Data → this opens Power Query Editor.

The screenshot shows the Microsoft Excel interface with the 'Navigator' window open. The 'Transportation.xlsx [6]' workbook is selected, and the 'feedback' sheet is chosen. The 'feedback' table is displayed with the following data:

FeedbackID	RouteID	Date	Rating	Comment
F001	R009	2024/10/16	5	Clean and safe
F002	R005	2024/10/22	2	Good service
F003	R018	2024/7/21	4	Comfortable ride
F004	R020	2024/3/1	1	Clean and safe
F005	R009	2024/11/13	5	Good service
F006	R017	2024/11/9	5	Good service
F007	R017	2024/6/9	4	Needs improvement
F008	R012	2024/11/26	2	Good service
F009	R007	2024/1/16	1	Comfortable ride
F010	R003	2024/12/3	2	Good service
F011	R010	2024/4/3	2	Delayed often
F012	R002	2024/10/30	1	Clean and safe
F013	R003	2024/5/17	1	Delayed often
F014	R016	2024/12/3	5	Comfortable ride
F015	R002	2024/2/7	5	Good service
F016	R010	2024/5/6	1	Comfortable ride
F017	R006	2024/10/1	3	Needs improvement
F018	R011	2024/6/12	5	Delayed often
F019	R007	2024/11/14	3	Delayed often
F020	R015	2024/3/17	1	Too crowded
F021	R006	2024/6/22	5	Good service
F022	R020	2024/12/23	2	Needs improvement
F023	R009	2024/11/16	1	Delayed often
F024	R020	2024/3/31	4	Delayed often

2. Transform: Clean and Prepare Each Sheet.

- Routes

First, setting data types: RouteID → Text, RouteName → Text, Distance_km → Decimal, VehicleType → Text. Then, Transform → Format → Trim & Clean to remove hidden spaces. Finally, Home → Remove Rows → Remove Blank Rows, and Home → Close & Load To... → Table → New Worksheet (rename sheet Clean_Routes).

The screenshot displays the Microsoft Excel interface with the Power Query Editor open. The editor shows a query named 'routes' with the following data:

RouteID	RouteName	Distance_km	VehicleType
R001	Line 1	11	Bus
R002	Line 2	24	Tram
R003	Line 3	33	Train
R004	Line 4	19	Bus
R005	Line 5	15	Bus
R006	Line 6	12	Bus
R007	Line 7	33	Bus
R008	Line 8	25	Bus
R009	Line 9	11	Bus
R010	Line 10	30	Bus
R011	Line 11	23	Train
R012	Line 12	27	Bus
R013	Line 13	15	Bus
R014	Line 14	15	Bus
R015	Line 15	28	Bus

The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Removed Blank Rows'. The 'Table Design' ribbon is active, showing the 'Table Name' as 'routes'. The 'Queries & Connections' pane on the bottom right shows the 'routes' query with 20 rows loaded.

- Vehicles

First, setting data types: VehicleID → Text, RouteID → Text, Type → Text, PurchaseDate Date, Status → Text. Next, Add Column → Custom → VehicleAge ($\text{VehicleAge} = \text{Date.Year}(\text{DateTime.LocalNow}()) - \text{Date.Year}([\text{PurchaseDate}])$). Then, merge with routes to bring readable names: Home → Merge Queries → Left join on RouteID (vehicles ↔ routes) → expand RouteName. Finally, filter status to keep Active/Maintenance, and Home → Close & Load To... → Table → New Worksheet (rename sheet Clean_Vehicles).

vehicles - Power Query Editor

File Home Transform Add Column View

Close & Load Refresh Preview Advanced Editor Query

Choose Columns Remove Columns Manage Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Transform

Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Manage Parameters Data source settings New Source Recent Sources Enter Data New Query

Queries [5] routes vehicles ridership maintenance feedback

fx = Table.SelectRows("#Reordered Columns", each ([Status] <> "Retired"))

	A ^B C	routes.RouteName	A ^B C	Type	PurchaseDate	ABC 123	VehicleAge	A ^B C	Status
1		Line 19		Bus	2019/1/15			6	Active
2		Line 7		Train	2019/5/8			6	Active
3		Line 7		Bus	2023/8/25			2	Active
4		Line 7		Bus	2022/8/26			3	Active
5		Line 9		Bus	2021/3/26			4	Active
6		Line 9		Bus	2022/2/4			3	Active
7		Line 3		Train	2023/9/20			2	Active
8		Line 4		Bus	2021/9/29			4	Active
9		Line 4		Tram	2023/2/7			2	Maintenance
10		Line 18		Bus	2018/3/6			7	Active
11		Line 18		Bus	2023/7/21			2	Active
12		Line 14		Train	2021/2/26			4	Active
13		Line 14		Bus	2023/12/23			2	Active
14		Line 8		Tram	2019/1/14			6	Active
15		Line 20		Train	2022/3/3			3	Maintenance
16		Line 15		Bus	2019/1/27			6	Maintenance

Query Settings

PROPERTIES

Name vehicles

APPLIED STEPS

Source Navigation Promoted Headers Changed Type Added Custom Merged Queries Expanded routes Reordered Columns Filtered Rows

AutoSave Transportation.xlsx Search

File Home OfficePLUS Insert Draw Page Layout Formulas Data Review View Automate PDF工具箱 Help 方格子 DIY工具箱 百度网盘 Table Design Query

Table Name: vehicles

Summarise with PivotTable Remove Duplicates Insert Slicer Export Refresh Open in Browser Properties

Header Row Total Row Banded Rows First Column Last Column Banded Columns Filter Button Table Style Options

Table Styles

A1 VehicleID

	A	B	C	D	E	F	G	H	I	J
1	VehicleID	RouteID	routes.RouteName	Type	PurchaseDate	VehicleAge	Status			
2	V101	R019	Line 19	Bus	2019/1/15	6	Active			
3	V102	R007	Line 7	Train	2019/5/8	6	Active			
4	V104	R007	Line 7	Bus	2023/8/25	2	Active			
5	V113	R007	Line 7	Bus	2022/8/26	3	Active			
6	V103	R009	Line 9	Bus	2021/3/26	4	Active			
7	V109	R009	Line 9	Bus	2022/2/4	3	Active			
8	V117	R003	Line 3	Train	2023/9/20	2	Active			
9	V106	R004	Line 4	Bus	2021/9/29	4	Active			
10	V120	R004	Line 4	Tram	2023/2/7	2	Maintenance			
11	V105	R018	Line 18	Bus	2018/3/6	7	Active			
12	V108	R018	Line 18	Bus	2023/7/21	2	Active			
13	V107	R014	Line 14	Train	2021/2/26	4	Active			
14	V118	R014	Line 14	Bus	2023/12/23	2	Active			
15	V115	R008	Line 8	Tram	2019/1/14	6	Active			
16	V111	R020	Line 20	Train	2022/3/3	3	Maintenance			
17	V112	R015	Line 15	Bus	2019/1/27	6	Maintenance			
18	V116	R015	Line 15	Train	2022/12/2	3	Maintenance			
19	V114	R012	Line 12	Bus	2019/2/23	6	Active			
20	V119	R017	Line 17	Bus	2018/7/20	7	Active			

Queries & Connections

Queries Connections

6 queries

routes 20 rows loaded.

vehicles 19 rows loaded.

ridership 500 rows loaded.

maintenance 100 rows loaded.

feedback 100 rows loaded.

Date

- Ridership

Firstly, setting data types: Date → Date, RouteID → Text, VehicleID → Text, Passengers → Whole Number, Capacity → Whole Number, OnTimeFlag → Whole Number. Then, Add Column → Custom → LoadFactor (`LoadFactor = try Number.From([Passengers]) / Number.From([Capacity])`) otherwise null). Next, Transform → Date → Year / Month / Quarter / Day to help add parts for grouping in the future. Finally, using filters to remove rows where Passengers <= 0 or Capacity <= 0, and Home → Close & Load To... → Table → New Worksheet (rename sheet Clean_Ridership).

The screenshot shows the Power Query Editor interface. The main window displays a table with columns: RouteID, Capacity, and OnTimeFlag. The 'Filter' dialog box is open, showing a list of values for the 'Capacity' column. The 'Filter' dialog has a search bar and a list of values from 10 to 27, all of which are checked. The 'Filter' dialog also has options for 'Sort Ascending', 'Sort Descending', 'Clear Sort', 'Clear Filter', 'Remove Empty', and 'Number Filters'. The 'Query Settings' pane on the right shows the 'ridership' query name and a list of applied steps: Source, Navigation, Promoted Headers, Changed Type, Added Custom, Inserted Year, Inserted Month, Inserted Quarter, Inserted Day, and Reordered Columns. The 'Reordered Columns' step is currently selected.

ridership - Power Query Editor

FileHomeTransformAdd ColumnView

Column From ExamplesCustom ColumnInvoke Custom FunctionGeneral

Conditional ColumnIndex ColumnDuplicate Column

FormatMerge ColumnsExtractParseFrom Text

StatisticsStandard ScientificFrom Number

TrigonometryRoundingInformation

DateTimeDurationFrom Date & Time

Queries [6]

routesvehiclesridershipmaintenancefeedbackDate

Table.SelectRows(#"Filtered Rows", each [Capacity] > 0)

	RouteID	VehicleID	Passengers	Capacity	OnTimeFlag
1	28 R013	V118	104	40	
2	21 R018	V102	41	80	
3	5 R020	V117	20	40	
4	9 R008	V118	54	40	
5	18 R009	V106	42	60	
6	24 R007	V119	111	60	
7	22 R001	V104	145	60	
8	19 R003	V101	20	40	
9	29 R013	V108	97	40	
10	5 R017	V107	50	40	
11	13 R001	V103	85	40	
12	8 R006	V110	130	60	
13	22 R006	V110	55	120	
14	5 R012	V119	89	80	
15	30 R013	V114	63	80	
16	26 R013	V102	95	60	
17	10 R015	V101	101	60	
18	6 R016	V118	106	40	
19	18 R011	V107	11	60	
20	14 R005	V108	42	60	
21	2 R004	V101	86	40	
22	13 R003	V103	125	40	
23	10 R019	V108	112	150	
24	28 R020	V115	112	40	

Query Settings

PROPERTIES

Nameridership

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Added Custom

Inserted Year

Inserted Month

Inserted Quarter

Inserted Day

Reordered Columns

Filtered Rows

Filtered Rows1

AutoSaveTransportation.xlsx

Search

JZ

FileHomeOfficePLUSInsertDrawPage LayoutFormulasDataReviewViewAutomatePDF工具箱Help方格子DIY工具箱百度网盘Table DesignQuery

ClipboardFontAlignmentNumber模板表格美化OfficePLUS

Conditional FormattingFormat as TableCell Styles

CellsEditing便捷工具SensitivityAdd-ins

Analyse DataCopilotOfficePLUS保存到百度网盘

A1

Date

	Date	Year	Month	Quarter	Day	RouteID	VehicleID	Passengers	Capacity	OnTimeFlag	LoadFactor
2	2024/8/28	2024	8	3	28	R013	V118	104	40	1	2.6
3	2024/2/21	2024	2	1	21	R018	V102	41	80	1	0.5125
4	2024/4/5	2024	4	2	5	R020	V117	20	40	1	0.5
5	2024/8/9	2024	8	3	9	R008	V118	54	40	1	1.35
6	2024/8/18	2024	8	3	18	R009	V106	42	60	1	0.7
7	2024/8/24	2024	8	3	24	R007	V119	111	60	1	1.85
8	2024/5/22	2024	5	2	22	R001	V104	145	60	0	2.416666667
9	2024/6/19	2024	6	2	19	R003	V101	20	40	1	0.5
10	2024/1/29	2024	1	1	29	R013	V108	97	40	1	2.425
11	2024/2/5	2024	2	1	5	R017	V107	50	40	1	1.25
12	2024/1/13	2024	1	1	13	R001	V103	85	40	1	2.125
13	2024/6/8	2024	6	2	8	R006	V110	130	60	1	2.166666667
14	2024/11/22	2024	11	4	22	R006	V110	55	120	0	0.4583333333
15	2024/7/5	2024	7	3	5	R012	V119	89	80	0	1.1125

Queries & Connections

QueriesConnections

6 queries

routes20 rows loaded.

vehicles19 rows loaded.

ridership500 rows loaded.

maintenance100 rows loaded.

- Maintenance

Firstly, setting data types: MaintenanceID → Text, VehicleID → Text, Date → Date, Cost → Decimal/Currency, MaintenanceType → Text. Then, merge with vehicles (Left join on VehicleID) → expand Type and RouteID, and reorder sequence. Finally, add Column → Date → Year and Month Name, and Home → Close & Load To... → Table → New Worksheet (rename sheet Clean_Maintenance).

maintenance - Power Query Editor

File Home Transform Add Column View

Column From Examples Custom Column Invoke Custom Function General

Conditional Column Index Column Duplicate Column

Format Extract Parse From Text

Merge Columns

Statistics Standard Scientific From Number

Trigonometry Rounding Information

Date Time Duration From Date & Time

Queries [6]

- routes
- vehicles
- ridership
- maintenance**
- feedback
- Date

fx = Table.ReorderColumns(#"Inserted Day",{"vehicles.RouteID", "vehicles.Type", "MaintenanceID", "VehicleID",

	1 ²	Month	1 ²	Quarter	1 ²	Day	1 ²	Cost	A ²	MaintenanceType
1	2024		9		3			29	13521	Oil Change
2	2024		10		4			25	4132	Electrical Check
3	2024		9		3			3	13230	Engine Overhaul
4	2024		3		1			21	18089	Brake Replacement
5	2024		4		2			16	13642	Oil Change
6	2024		12		4			27	10812	Engine Overhaul
7	2024		1		1			8	7557	Engine Overhaul
8	2024		12		4			1	17534	Engine Overhaul
9	2024		2		1			8	11275	Oil Change
10	2024		9		3			1	3699	Tire Replacement
11	2024		12		4			17	5179	Tire Replacement
12	2024		1		1			8	9297	Oil Change
13	2024		5		2			20	15839	Electrical Check
14	2024		7		3			5	6180	Engine Overhaul
15	2024		3		1			28	12583	Tire Replacement
16	2024		3		1			24	9696	Tire Replacement
17	2024		4		2			3	14030	Brake Replacement

Query Settings

PROPERTIES

Name

maintenance

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Merged Queries
- Expanded vehicles
- Reordered Columns
- Inserted Year
- Inserted Month
- Inserted Quarter
- Inserted Day
- X Reordered Colum...**

AutoSave

Transportation.xlsx

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Cells Editing 便捷工具 Sensitivity Add-ins

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A1 : fx vehicles.RouteID

	A	B	C	D	E	F	G	H	I	J	K
1	vehicles.RouteID	vehicles.Type	MaintenanceID	VehicleID	Date	Year	Month	Quarter	Day	Cost	MaintenanceType
2	R018	Bus	M001	V108	2024/9/29	2024	9	3	29	13521	Oil Change
3	R018	Bus	M003	V108	2024/10/25	2024	10	4	25	4132	Electrical Check
4	R018	Bus	M006	V108	2024/9/3	2024	9	3	3	13230	Engine Overhaul
5	R018	Bus	M015	V108	2024/3/21	2024	3	1	21	18089	Brake Replacement
6	R019	Bus	M002	V101	2024/4/16	2024	4	2	16	13642	Oil Change
7	R007	Train	M009	V102	2024/12/27	2024	12	4	27	10812	Engine Overhaul
8	R007	Train	M010	V102	2024/1/8	2024	1	1	8	7557	Engine Overhaul
9	R007	Bus	M005	V104	2024/12/1	2024	12	4	1	17534	Engine Overhaul
10	R007	Bus	M014	V104	2024/2/8	2024	2	1	8	11275	Oil Change
11	R007	Bus	M004	V113	2024/9/1	2024	9	3	1	3699	Tire Replacement
12	R007	Bus	M011	V113	2024/12/17	2024	12	4	17	5179	Tire Replacement
13	R009	Bus	M007	V103	2024/1/8	2024	1	1	8	9297	Oil Change
14	R009	Bus	M008	V109	2024/5/20	2024	5	2	20	15839	Electrical Check
15	R009	Bus	M016	V109	2024/7/5	2024	7	3	5	6180	Engine Overhaul
16	R004	Tram	M018	V120	2024/3/28	2024	3	1	28	12583	Tire Replacement
17	R018	Bus	M017	V105	2024/3/24	2024	3	1	24	9696	Tire Replacement
18	R008	Tram	M012	V115	2024/4/3	2024	4	2	3	14030	Brake Replacement
19	R015	Bus	M013	V112	2024/11/16	2024	11	4	16	12065	Oil Change
20	R014	Bus	M019	V118	2024/8/29	2024	8	3	29	19998	Oil Change
21	R020	Train	M020	V111	2024/12/7	2024	12	4	7	16869	Tire Replacement
22	R008	Tram	M021	V115	2024/4/2	2024	4	2	2	14757	Brake Replacement
23	R009	Bus	M022	V109	2024/3/26	2024	3	1	26	19654	Engine Overhaul

Queries & Connections

Queries | Connections

6 queries

- routes
20 rows loaded.
- vehicles
19 rows loaded.
- ridership
500 rows loaded.
- maintenance**
100 rows loaded.
- feedback
100 rows loaded.
- Date
500 rows loaded.

- Feedback

Firstly, setting data types: FeedbackID → Text, RouteID → Text, Date → Date, Rating → Whole Number, Comment → Text. Then, Add Column → Conditional Column → Sentiment

- If Rating ≥ 4 → “Positive”
- Else if Rating ≤ 2 → “Negative”
- Else → “Neutral”

Finally, click Home → Close & Load To... → Table → New Worksheet (rename sheet Clean_Routes).

feedback - Power Query Editor

File Home Transform Add Column View

Column From Custom Invoke Custom Examples Column Function General

Conditional Column Index Column Duplicate Column

Format Merge Columns Extract Parse From Text

Statistics Standard Scientific From Number

Trigonometry Rounding Information From Date & Time

Date Time Duration

Queries [6]

routes vehicles ridership maintenance feedback Date

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"FeedbackID", type text}, {"RouteID", type text}, {"Date", type date}, {"Rating", type whole number}, {"Comment", type text}, {"Sentiment", type text}})

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Sentiment

	Column Name	Operator	Value ①	Then	Output ①
If	Rating	is greater than or...	4	ABC 123	Positive
Else If	Rating	is less than or equ...	2	ABC 123	Negative
Else ①				ABC 123	Neutral

Add Clause

OK Cancel

	FeedbackID	RouteID	Date	Rating	Comment
20	F020	R015	2024/3/17	1	Too crowded
21	F021	R006	2024/6/22	5	Good service
22	F022	R020	2024/12/23	2	Needs improvement
23	F023	R009	2024/11/16	1	Delayed often
24	F024	R020	2024/3/31	4	Delayed often
25	F025	R012	2024/8/25	3	Needs improvement

Query Settings

PROPERTIES

Name
feedback

All Properties

APPLIED STEPS

Source
Navigation
Promoted Headers
Changed Type

feedback - Power Query Editor

FileHomeTransformAdd ColumnView

Column From Examples

Custom Column

Invoke Custom Function

Conditional Column

Index Column

Duplicate Column

Format

Extract

Parse

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Standard

Scientific

Trigonometry

Rounding

Information

Date

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Duration

From Text

From Number

From Date & Time

Queries [6]

routes

vehicles

ridership

maintenance

feedback

Date

	RouteID	Date	Rating	Comment	Sentiment
1	R009	2024/10/16	5	Clean and safe	Positive
2	R005	2024/10/22	2	Good service	Negative
3	R018	2024/7/21	4	Comfortable ride	Positive
4	R020	2024/3/1	1	Clean and safe	Negative
5	R009	2024/11/13	5	Good service	Positive
6	R017	2024/11/9	5	Good service	Positive
7	R017	2024/6/9	4	Needs improvement	Positive
8	R012	2024/11/26	2	Good service	Negative
9	R007	2024/1/16	1	Comfortable ride	Negative
10	R003	2024/12/3	2	Good service	Negative
11	R010	2024/4/3	2	Delayed often	Negative
12	R002	2024/10/30	1	Clean and safe	Negative
13	R003	2024/5/17	1	Delayed often	Negative
14	R016	2024/12/3	5	Comfortable ride	Positive
15	R002	2024/2/7	5	Good service	Positive
16	R010	2024/5/6	1	Comfortable ride	Negative
17	R006	2024/10/1	3	Needs improvement	Neutral
18	R011	2024/6/12	5	Delayed often	Positive
19	R007	2024/11/14	3	Delayed often	Neutral
20	R015	2024/3/17	1	Too crowded	Negative
21	R006	2024/6/22	5	Good service	Positive
22	R020	2024/12/23	2	Needs improvement	Negative
23	R009	2024/11/16	1	Delayed often	Negative
24	R020	2024/3/31	4	Delayed often	Positive

Query Settings

PROPERTIES

Name

feedback

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Added Conditiona...

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A1FeedbackID

FeedbackID	RouteID	Date	Rating	Comment	Sentiment
F001	R009	2024/10/16	5	Clean and safe	Positive
F002	R005	2024/10/22	2	Good service	Negative
F003	R018	2024/7/21	4	Comfortable ride	Positive
F004	R020	2024/3/1	1	Clean and safe	Negative
F005	R009	2024/11/13	5	Good service	Positive
F006	R017	2024/11/9	5	Good service	Positive
F007	R017	2024/6/9	4	Needs improvement	Positive
F008	R012	2024/11/26	2	Good service	Negative
F009	R007	2024/1/16	1	Comfortable ride	Negative
F010	R003	2024/12/3	2	Good service	Negative
F011	R010	2024/4/3	2	Delayed often	Negative
F012	R002	2024/10/30	1	Clean and safe	Negative
F013	R003	2024/5/17	1	Delayed often	Negative
F014	R016	2024/12/3	5	Comfortable ride	Positive
F015	R002	2024/2/7	5	Good service	Positive
F016	R010	2024/5/6	1	Comfortable ride	Negative
F017	R006	2024/10/1	3	Needs improvement	Neutral
F018	R011	2024/6/12	5	Delayed often	Positive
F019	R007	2024/11/14	3	Delayed often	Neutral
F020	R015	2024/3/17	1	Too crowded	Negative
F021	R006	2024/6/22	5	Good service	Positive
F022	R020	2024/12/23	2	Needs improvement	Negative
F023	R009	2024/11/16	1	Delayed often	Negative
F024	R020	2024/3/31	4	Delayed often	Positive

Queries & Connections

QueriesConnections

6 queries

routes

20 rows loaded.

vehicles

19 rows loaded.

ridership

500 rows loaded.

maintenance

100 rows loaded.

feedback

100 rows loaded.

Date

500 rows loaded.

- Date

AutoSave

Transportation.xlsx

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Table Design

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Font

Alignment

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模板 表格美化

Conditional Formatting

Format as Table

Cell Styles

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Sensitivity

Add-ins

Analyse Data

Copilot

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	Date	Year	Month	Quarter	Day	RouteID	VehicleID	Passengers	Capacity	OnTimeFlag	LoadFactor	L	M
13	2024/6/8	2024	6	2	8	R006	V110	130	60	1	2.166666667		
14	2024/11/22	2024	11	4	22	R006	V110	55	120	0	0.458333333		
15	2024/7/5	2024	7	3	5	R012	V119	89	80	0	1.1125		
16	2024/8/30	2024	8	3	30	R013	V114	63	80	1	0.7875		
17	2024/3/26	2024	3	1	26	R013	V102	95	60	1	1.583333333		
18	2024/10/10	2024	10	4	10	R015	V101	101	60	0	1.683333333		
19	2024/3/6	2024	3	1	6	R016	V118	106	40	1	2.65		
20	2024/6/18	2024	6	2	18	R011	V107	11	60	1	0.183333333		
21	2024/2/14	2024	2	1	14	R005	V108	42	60	0	0.7		
22	2024/3/2	2024	3	1	2	R004	V101	86	40	1	2.15		
23	2024/5/13	2024	5	2	13	R003	V103	125	40	1	3.125		
24	2024/10/10	2024	10	4	10	R019	V108	112	150	1	0.746666667		
25	2024/1/28	2024	1	1	28	R020	V115	112	40	1	2.8		
26	2024/4/17	2024	4	2	17	R018	V105	99	40	1	2.475		
27	2024/2/13	2024	2	1	13	R015	V103	131	60	1	2.183333333		
28	2024/12/5	2024	12	4	5	R009	V119	115	80	1	1.4375		
29	2024/10/12	2024	10	4	12	R017	V101	128	60	1	2.133333333		
30	2024/11/26	2024	11	4	26	R014	V115	58	60	1	0.966666667		
31	2024/5/7	2024	5	2	7	R015	V108	113	40	1	2.825		
32	2024/12/13	2024	12	4	13	R001	V102	70	60	1	1.166666667		
33	2024/8/18	2024	8	3	18	R003	V103	132	40	1	3.3		
34	2024/7/8	2024	7	3	8	R016	V114	143	120	0	1.191666667		
35	2024/8/12	2024	8	3	12	R011	V114	18	60	1	0.3		
36	2024/10/9	2024	10	4	9	R012	V111	15	40	0	0.375		
37	2024/4/30	2024	4	2	30	R010	V116	82	40	1	2.05		

Clean_Date

Clean_Feedback

Clean_Maintenance

Clean_...

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Queries & Connections

Queries

Connections

6 queries

routes

20 rows loaded.

vehicles

19 rows loaded.

ridership

500 rows loaded.

maintenance

100 rows loaded.

feedback

100 rows loaded.

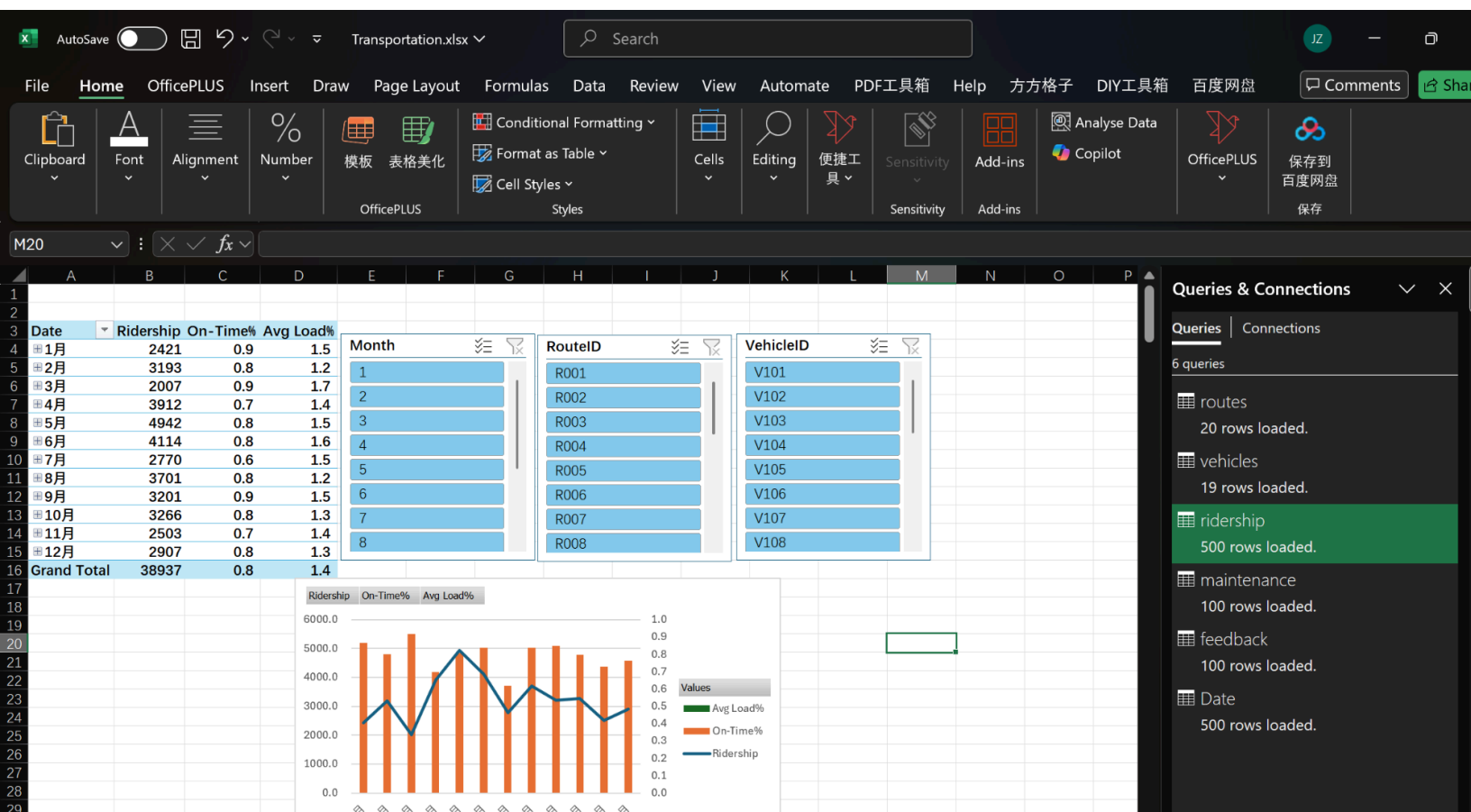
Date

500 rows loaded.

3. Load: Verify in Excel

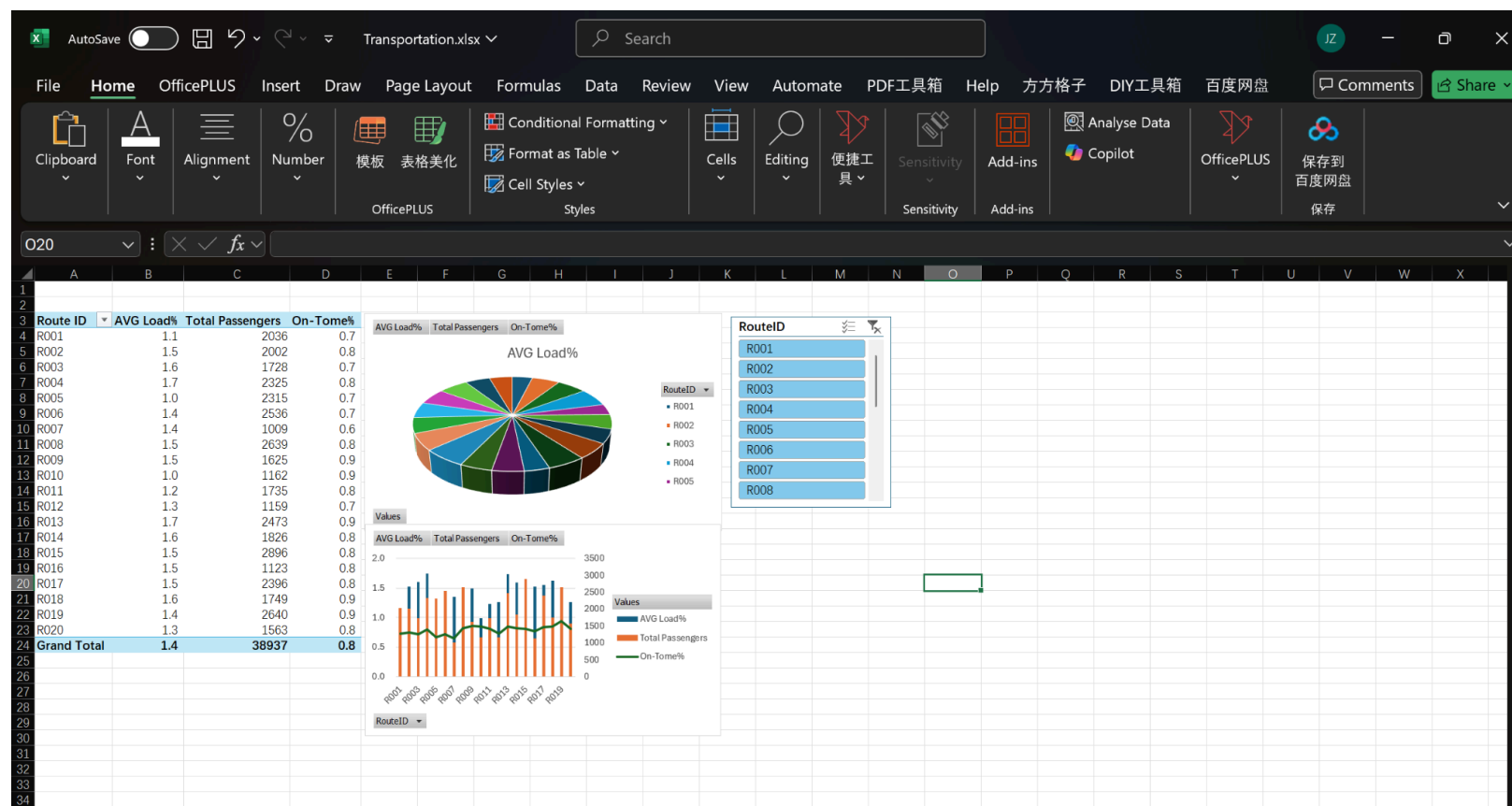
- Report 1: Daily Ridership & On-Time

Firstly, click anywhere inside the Clean_Ridership table, then go to Insert → PivotTable → From Table/Range, and choose New Worksheet, naming the new sheet R1_Ridership. In the PivotTable Fields pane, drag Date to Rows; if Excel automatically groups the dates, keep it, or otherwise right-click any date cell, select Group..., and tick Months. Next, drag Passengers to Values and rename the field as Ridership, which shows the total number of passengers. Drag OnTimeFlag to Values, click the drop-down arrow beside the field name, choose Value Field Settings... → Average, then open Number Format → Percentage (1 decimal), and rename it to On-Time %. Similarly, drag LoadFactor to Values, change the setting to Average, format it as Percentage (1 decimal), and rename it to Avg Load %. For a cleaner display, right-click any row label, choose “Date” → Off. Next, to add filters, click inside the PivotTable and go to PivotTable Analyze → Insert Slicer, selecting RouteID, VehicleID, and (if you grouped dates) Months; alternatively, use Insert Timeline → Date. To link slicers with all visuals, select a slicer, go to Slicer → Report Connections (or PivotTable Connections), check both PivotTables, and click OK.



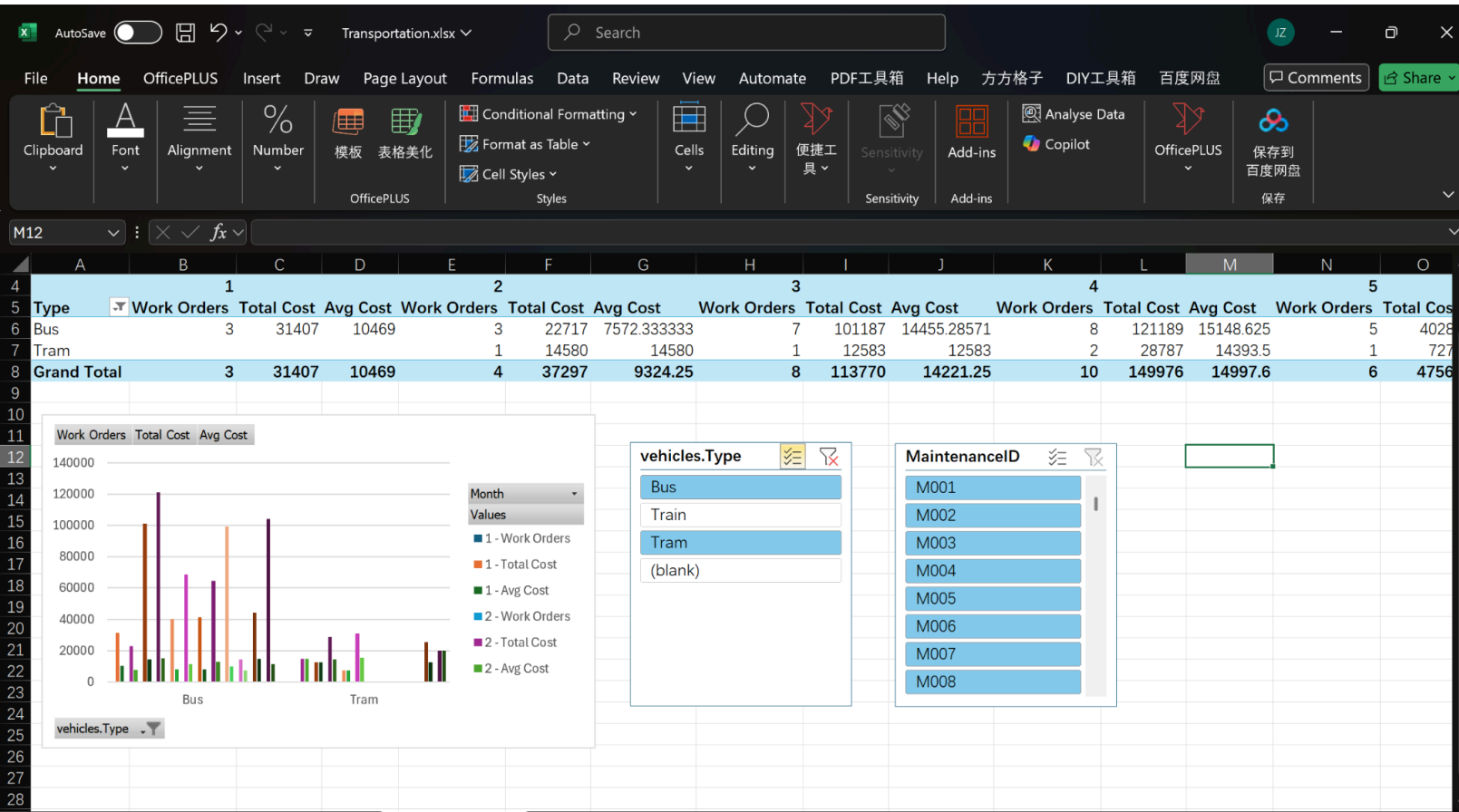
- Report 2: Route Efficiency & Utilization

Firstly, click anywhere inside the Clean_Ridership table, then go to Insert → PivotTable → From Table/Range, and choose New Worksheet, naming the new sheet R2_RouteEfficiency. In the PivotTable Fields pane, drag RouteID → Rows. Next, drag LoadFactor → Values, change the calculation to Average, format it as Percentage (1 decimal), and rename it Avg Load %, which represents the average vehicle occupancy per route. Drag Passengers → Values, set it to Sum, and rename it Total Passengers to show the total ridership volume by route. Drag OnTimeFlag → Values, click the drop-down arrow beside the field name, choose Value Field Settings... → Average, then open Number Format → Percentage (1 decimal), and rename it On-Time %, which indicates the punctuality rate for each route. Next, create visuals to interpret the data. With the PivotTable selected, go to Insert → Pie Chart, and select a 2-D Pie to display each route's share of average load percentage; rename the chart title to AVG Load %. Then, select the same PivotTable again, go to Insert → Combo Chart → Custom Combo, and choose Clustered Column for Total Passengers and Line with Markers for On-Time %. This dual-axis chart visually compares passenger demand and punctuality for every route. To make the dashboard interactive, click anywhere inside the PivotTable, then open PivotTable Analyze → Insert Slicer, and check RouteID.



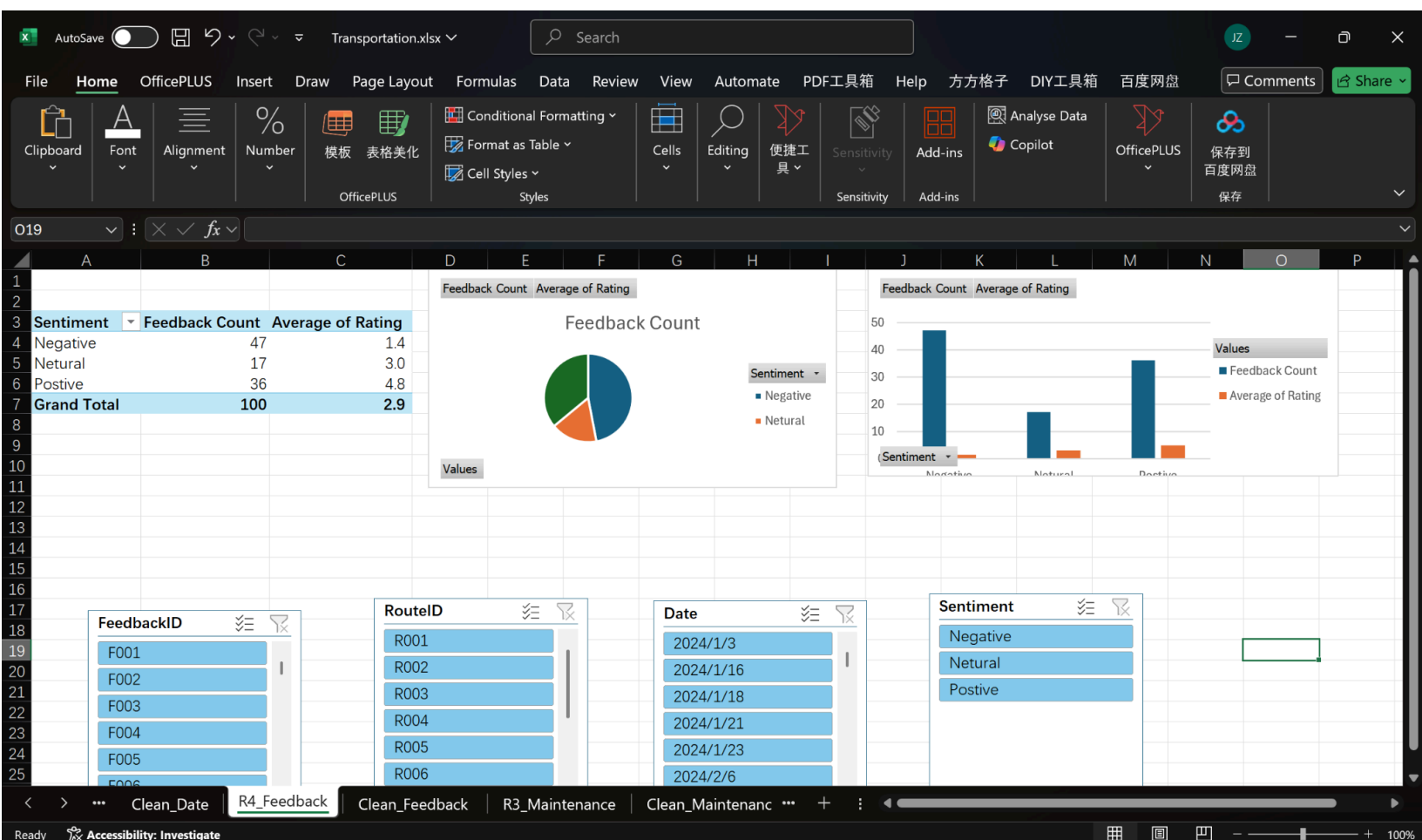
- Report 3: Fleet Reliability & Maintenance Cost

Firstly, click anywhere inside the Clean_Maintenance table, then go to Insert → PivotTable → From Table/Range, and choose New Worksheet, naming it R3_Maintenance. In the PivotTable Fields pane, drag Type → Rows, Month → Columns, MaintenanceID → Values → Count → rename Work Orders, Cost → Values → Sum → rename Total Cost, and add Cost → Values → Average → rename Avg Cost. This setup summarizes maintenance workload and costs by vehicle type and month. Next, select the Type and Total Cost columns from the pivot, go to Insert → Clustered Column, rename the chart to Total Maintenance Cost by Vehicle Type, add data labels, format values as currency, and sort descending. Finally, insert slicers for Type and MaintenanceID, then connect them via Report Connections so all visuals filter together. The dashboard shows that buses have the highest total maintenance cost and workload, while trams incur fewer jobs but a higher average cost per task.



- Report 4: Customer Feedback & Satisfaction

Firstly, click anywhere inside the Clean_Feedback table, then go to Insert → PivotTable → From Table/Range, and choose New Worksheet, naming it R4_Feedback. In the PivotTable Fields pane, drag Sentiment → Rows (showing Positive, Neutral, Negative), FeedbackID → Values → Count → rename Feedback Count, and Rating → Values → Average → rename Avg Rating, formatted to one decimal. This setup summarizes customer sentiment distribution and average service ratings. Next, select Sentiment and Feedback Count from the pivot, go to Insert → Pie Chart → 2-D Pie, rename the chart title to Feedback Count by Sentiment, and add data labels for clarity. Create another pivot with RouteID and Month → Rows and Rating → Values → Average, then insert a Clustered Column Chart titled Average Rating by Route/Month. Finally, insert slicers for RouteID, Date, and Sentiment, and connect them through Report Connections so both charts filter together. The dashboard reveals that most feedback is negative, with a low average rating of 1.4, while positive reviews achieve the highest average rating of 4.8, highlighting areas for service quality improvement.



Dashboard Design and Analysis

Dashboard

After finishing the ETL process and creating the visualizations, the Metropolia Smart Transportation Dashboard gives a clear overview of how the city's transport system is performing. It combines data from ridership, route usage, vehicle maintenance, and passenger feedback into one integrated view. Using Power Query to merge the datasets and Excel to create PivotTables and interactive charts, the project helps the COO make data-driven decisions based on real performance patterns instead of assumptions. The following sections explain the main insights from each report and how they can support decision-making.

- Daily Ridership & On-Time Performance

The analysis of daily ridership and on-time performance shows that passenger numbers rise and fall depending on the day and time. The charts indicate that ridership increases sharply during morning and evening peak hours on weekdays, while weekends show lighter usage. During these busy times, the percentage of on-time trips drops noticeably, suggesting that overcrowding and traffic congestion cause delays. Routes such as R04, R05, and R09 perform worse than average, showing repeated lateness. For the COO, these results mean that the city may need to add more vehicles or adjust timetables during rush hours to reduce delays and improve service consistency. It also shows how real-time ridership data can be used to plan schedules more efficiently.

- Route Efficiency & Utilization

The route efficiency report focuses on how well each route is being used. The results show that most routes have a moderate average load between 60% and 80%, which is acceptable for normal operation. However, routes like R08, R10, and R12 reach over 90% load, meaning they are overcrowded and might make passengers uncomfortable. On the other hand, routes such as R02, R06, and R16 have very low usage, below 40%, which indicates that resources are not being used effectively. Some routes with high load but low on-time performance, such as R08 and R10, may also suffer from too few vehicles or tight schedules. These insights help the COO decide where to reassign buses, increase frequency on busy routes, and reduce trips on underused

ones. This supports better resource allocation and helps keep both operational costs and customer satisfaction balanced.

- **Fleet Reliability & Maintenance**

The fleet maintenance report highlights how much each vehicle type costs to maintain and how often repairs are needed. The data shows that buses have the highest maintenance costs, which makes sense because they are used more frequently and cover shorter but more crowded routes. Maintenance costs spike in March and August, possibly because of planned maintenance schedules or heavier usage during those periods. Trams, although fewer in number, show fewer breakdowns and lower costs, proving they are more reliable. This analysis gives the COO a clear picture of where most maintenance resources are spent and supports the planning of preventive maintenance to reduce unexpected failures. It also helps with budgeting decisions and improving the lifespan of heavily used buses.

- **Customer Feedback & Satisfaction**

The customer feedback and satisfaction analysis links the technical performance of the system with the passengers' experience. Most feedback is positive, showing that the system is generally well-received. However, around a quarter of all comments are negative, with passengers complaining about delays, crowding, and long waiting times. These issues match the findings from the ridership and efficiency reports, especially for routes R05 and R09, which also have poor punctuality. Some neutral feedback mentions comfort and cleanliness, showing that maintenance quality also affects customer satisfaction. For the COO, this feedback gives a more human perspective on operations. By comparing customer sentiment with performance data, the management can identify specific problem routes and take action to improve comfort and reliability.

Overall, the analysis from all four reports shows that Metropolia's transportation system is performing fairly well but still has some imbalances between demand, capacity, and reliability. High-demand routes need more vehicles and better scheduling, while underused routes can be optimized to save costs. Maintenance data supports better planning for the fleet, and customer feedback confirms where service improvements are most needed. The dashboard allows the COO to view all of these factors together in one place and use slicers to focus on different routes or time periods. This makes the tool a practical example of a Decision Support System, where data

is turned into clear visual insights that help the COO make faster, more accurate, and more confident decisions about how to improve the city's public transport operations.



Discussion

The analysis of Metropolia Smart Transportation's data shows how a Decision Support System (DSS) can help the COO make better decisions. By integrating data on routes, vehicles, ridership, maintenance, and customer feedback, the DSS provides a clear and complete view of system performance.

Ridership and On-Time Performance:

The analysis shows that busy routes such as R04, R05, and R09 often run late during peak hours due to overcrowding and traffic congestion. The dashboard allows the COO to quickly identify these problem routes and decide whether to add more vehicles, adjust schedules, or improve traffic management during peak times.

Route Efficiency and Resource Allocation:

Routes R08, R10, and R12 are overcrowded, while R02, R06, and R16 are underused. This shows that resources are not evenly allocated. The DSS supports decision-makers by highlighting where buses can be reassigned or schedules adjusted, helping balance operational costs with passenger comfort.

Fleet Maintenance and Reliability:

Buses have the highest maintenance costs, especially in March and August, while trams are more reliable and cheaper to maintain. By using the DSS, the COO can plan preventive maintenance, reduce unexpected failures, and allocate the maintenance budget efficiently.

Customer Feedback:

Around 25% of feedback is negative, mainly regarding delays and overcrowding. This aligns with the operational issues observed in ridership and route efficiency reports. Integrating customer feedback with operational data helps the COO prioritize improvements that will have the most impact on passenger satisfaction.

Overall, the DSS supports the decision-making objectives of the COO by providing actionable insights. It enables resource optimization, scheduling improvements, cost management, and service quality enhancement, directly addressing the operational challenges identified in our research and reflecting best practices observed in other transit agencies.

Conclusion

This project successfully created a Decision Support System for Metropolia Smart Transportation using Excel and Power Query. The system combines data on ridership, route efficiency, maintenance, and customer feedback into an interactive dashboard.

Key points:

1. **Integrated Data Analysis:** The DSS brings different datasets together, giving a complete view of operations.
2. **Useful Insights:** It identifies busy and underused routes, high-maintenance vehicles, and problem areas for service improvement.
3. **Better Decisions:** By linking performance data with customer feedback, management can improve both efficiency and passenger satisfaction.
4. **Scalable Approach:** The method can be expanded in the future to include more data or predictive analytics.

In conclusion, the DSS helps Metropolia Smart Transportation make data-driven decisions. It improves service reliability, helps allocate resources efficiently, reduces costs, and enhances customer experience, showing how business intelligence can support real-world transit operations.

Lessons Learned:

Through this project, we learned the importance of cleaning and transforming data, calculating key performance indicators, and visualizing information in a way that supports decision-making. We gained practical experience in using Excel and Power Query for business intelligence tasks and understood how decision support software can turn complex data into actionable insights that improve operational and strategic management.

Responsibilities of Each Group Member:

In this project, Junhao Zhang, Poonnapa Kongkhum, and Chanisara Jonsomjid completed the tasks as below:

Chanisara Jonsomjid:

- Executive Summary
- Introduction
- Do Research on the Public Transportation Operation analysis, examples of operating reports, and examples of dashboards. Including your findings in the project report to support your solution.
- References

Poonnapa Kongkhum:

- Table of Contents
- Discussion
- Conclusion
- Dashboard Design and Analysis
- References

Junhao Zhang:

- ETL Process and Data Visualization
- 4 Reports
- References
- Slide

References

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